

Collaborative Multi-Agent Deep Reinforcement Learning for Joint Task Offloading and Resource Allocation with Long Term Energy Control

Chenglong Yu^{1,2,3}, Wang Xing^{1,2,3}, Jinglin Shi^{1,2,3}, Yiqing Zhou^{1,2,3} and Ling Liu^{1,2,3}

¹State Key Lab of Processors, Institute of Computing Technology, CAS, Beijing, 100190, China

²University of Chinese Academy of Sciences, Beijing, 100049, China

³Beijing Key Laboratory of Mobile Computing and Pervasive Device, Beijing, 100080, China

Email: yuchenglong24@mails.ucas.ac.cn, {xingwang, sjl, zhouyiqing, liuling}@ict.ac.cn

Abstract—In mobile edge computing (MEC) enabled Industrial Internet of Things (IIoT), task offloading and resource allocation are always jointly optimized to achieve the best energy efficiency of IIoT terminals, which is important for MEC-IIoT. Existing research mainly focused on the instant energy consumption for the current task, ignoring the fact that the energy consumption is long term since energy is also required for subsequent tasks. Excessive energy consumption by the current task will affect the execution of subsequent tasks, leading to a degraded quality of service (QoS). Meanwhile, to solve the joint optimization problem, existing multi-agent deep reinforcement learning (MADRL) based methods face challenges like slow convergence. To tackle these problems, this paper proposes a long-term energy control enabled joint optimization scheme for the task offloading and resource allocation (LTE-JTORA). The main idea is to maximize the terminal energy efficiency while enabling the terminal to complete as many tasks as possible, so that the long-term energy control is realized. Then, an energy efficiency (EE) reward based collaborative MADRL (EE-CMADRL) is proposed to solve the NP-hard optimization problem. Different to MADRL where multiple agents work independently, EE-CMADRL is based on the Centralized Training Distributed Execution (CTDE) framework, where multiple agents can work collaboratively to train one EE reward enabled critic network. With more diverse data from multiple actor networks, EE-CMADRL can converge fast. Simulation results show that, compared to existing MADRL algorithms, the proposed EE-CMADRL improves the convergence speed by 71%. Compared to existing schemes with instant energy control, with the same time and energy constraints, the proposed LTE-JTORA scheme increases the number of completed tasks by 57%, and improves energy efficiency by 68%.

Index Terms—Multi-agent deep reinforcement learning, Industrial Internet of Things, Mobile edge computing, Energy efficiency, Resource allocation.

I. INTRODUCTION

The sixth-generation mobile network (6G) is expected to enable a deeper integration of wireless communication and human activities, giving rise to a series of potential emerging applications, such as the Industrial Internet of Things (IIoT) [1]. Mobile edge computing (MEC) is a key technology for IIoT. Compared to the centralized mobile cloud computing (MCC), MEC can keep industrial data in the factory by computing at edge servers, which is important to protect the

data [2], [3]. Moreover, MEC can achieve a lower transmission delay and energy consumption than MCC.

Consider a MEC enabled IIoT (MEC-IIoT) network with many mobile terminals (MTs) powered by batteries such as Automated Guided Vehicles (AGVs). Obviously, the terminal energy efficiency is of high importance [4]–[6]. It is impacted by various elements, including task offloading and resource allocation, which are coupled with each other. Recently, joint task offloading and resource allocation of MEC-IIoT has drawn a lot of attention [2], [7], [8], taking energy efficiency into account. For example, the energy consumed by the completed tasks and current task is restricted to not exceeding the battery energy in [7]. Although these works concern the terminal energy efficiency in IIoT, they focus on one single task which may use up all the energy. This is not always applicable in practice, where MTs are expected to execute a series of tasks with limited energy [9] [10]. Thus, it is necessary to introduce a long-term control of battery energy within the operational duration of these MTs.

Moreover, concerning the solution to the optimization problem of joint task offloading and resource allocation, various methods have been developed, such as the compressive sensing based algorithm [11] and the Dynamic Network Slicing and Resource Allocation algorithm [12]. Both methods take the acquisition of complete network statuses as a prerequisite. Therefore, they may not well adapt to large-scale dynamic IIoT networks, where the network status is expensive to acquire and difficult to generalize [1]. Moreover, model-based algorithms such as Successive Convex Approximation (SCA) and Block Coordinate Descent (BCD) are usually time-consuming and computationally intensive in a time-varying IIoT environment, where they would keep solving the problem as long as the environment changes [3], [8]. Without requirements for prior knowledge, multi-agent deep reinforcement learning (MADRL) has emerged as a promising solution for the joint optimization problem [1]. MADRL can adapt to complex IIoT scenarios by making different actions with its intelligences [3] [7]. Each agent consists of an actor network and a critic network. For example, in [3], the weighted sum of latency and energy consumption is minimized by optimizing offloading

decisions, terminal transmission power, and terminal computational resource allocation. Then a multi-agent Dueling deep Q-network based algorithm is proposed to solve the problem. However, it should be noted that in existing MADRL, multiple agents work independently to each other, resulting in a slow convergence speed [13].

This paper concerns the long-term energy efficiency of MEC-IIoT. Thus, the joint optimization of task offloading and resource allocation should be carried out with a constraint of the long-term energy efficiency. The challenge is how to model the long-term energy efficiency. Moreover, to solve the joint optimization problem, instead of existing MADRL methods where agents work independently, collaborated MADRL is desirable which could enhance the convergence speed. The main contributions of this paper are summarized as follows:

- We propose a long-term energy control enabled joint optimization of task offloading and resource allocation scheme (LTE-JTORA). The main idea is to encourage the terminal to complete as many tasks as possible, thus the long-term energy control can be realized. Note that energy allocation should be carried out for multiple tasks instead of one single task, which is modeled in the joint optimization problems.
- We propose an energy efficient (EE) reward-based collaborative multi-agent DRL approach (EE-CMADRL) to solve the NP-hard joint optimization problem. Based on the Centralized Training Distributed Execution (CTDE) framework [14], agents in EE-CMADRL can collaborate to train one shared EE reward enabled critic network. With more diverse data from multiple actor networks, EE-CMADRL can converge faster.
- We evaluate the performance of the proposed LTE-JTORA scheme with EE-CMADRL. First, the proposed EE-CMADRL algorithm can converge rapidly after around 50 iterations, which is approximately 71% faster than existing MADRL algorithms. Meanwhile, with the same time and energy constraints, the proposed LTE-JTORA with EE-CMADRL increases the number of completed tasks by 57% and improves the energy efficiency by 68% compared to existing short-term energy control with different MADRL algorithms.

II. SYSTEM MODEL

As shown in Fig. 1, consider a MEC-IIoT system consisting of one BS co-located with an MEC server and N MTs (e.g. AGVs) indexed as $N=1,2,\dots,N$. Assume that the BS and MEC server have sufficient energy and computation capacity while the MTs are with limited battery energy and computation capacity. The wireless communications between the BS and AGVs are carried out based on OFDM with N -subcarriers and a total bandwidth of NB [1]. Each AGV is required to perform a series of transportation tasks. During this process, tasks such as obstacle recognition are computationally intensive, which involves image processing. Due to the limited computational and battery capacity of AGVs, they are usually unable to

handle computationally intensive tasks within predefined deadlines [2]. Thus, AGVs would collaborate with MEC servers to complete these computationally intensive tasks via task offloading and resource allocation [1], [7], [8].

A. MEC-IIoT

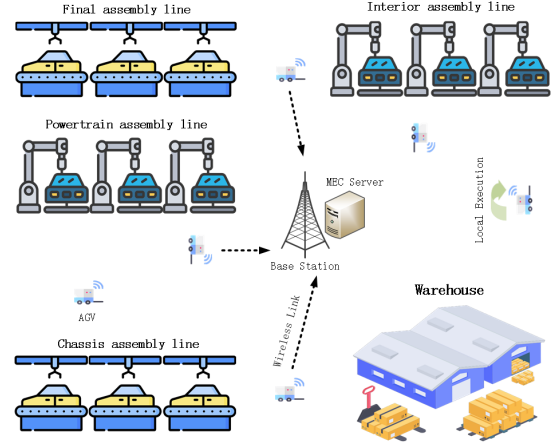


Fig. 1. An example MEC-IIoT network.

Denote a slot as Δt and consider a duration of $T = M\Delta t$. Assume that the autonomous driving tasks of AGVs are generated at the start of a slot. Let $TK_{n,m} = (L_{n,m}, \hat{T}_{n,m})$ denote the task generated by the AGV n at time m , where $L_{n,m}$ is the task size [7], [8] and $\hat{T}_{n,m}$ is the maximum tolerable delay of this task. Assume that tasks can be processed at each AGV or fully offloaded to the MEC server, but cannot be partially processed or offloaded [8]. Hence, denote the offloading decision of $TK_{n,m}$ as $x_{n,m}$, where $x_{n,m} = 1$ indicates local processing while $x_{n,m} = 0$ indicates offloading to MEC server.

B. Task Offloading Model

If the task $TK_{n,m}$ is offloaded to the MEC server, the calculation results should be returned to the AGV n . Considering scenarios that the size of the returned message is sufficiently small, the energy consumption and transmission delay of returning the results can be ignored.

To obtain the transmission delay of uploading the task $TK_{n,m}$ to the MEC server, the achievable uplink transmission rate for $TK_{n,m}$ can be given by

$$R_{n,m} = B \log_2 \left(1 + \frac{P_{n,m} h_{n,m}}{\sigma^2} \right), \quad (1)$$

where $P_{n,m}$ is the power of transmitting $TK_{n,m}$ to the MEC server, $h_{n,m}$ denotes the channel coefficient, and σ^2 is the noise variance. Thus the transmission delay of $TK_{n,m}$ in the uplink can be given by $T_{n,m}^{comm} = x_{n,m} L_{n,m} / R_{n,m}$. The transmission energy consumption of $TK_{n,m}$ is given by $E_{n,m}^{comm} = P_{n,m} T_{n,m}^{comm}$. The computation time on the MEC server is given by $T_{n,m}^{pro} = x_{n,m} L_{n,m} e_{n,m} / f^e$, where $e_{n,m}$ denotes the CPU cycles required to process each bit of data, and f^e denotes the CPU frequency of the MEC server.

C. Local Computing Model

If the task $TK_{n,m}$ is locally processed at the AGV n , the local computation delay of $TK_{n,m}$ can be given by

$$T_{n,m}^l = \frac{(1 - x_{n,m}) L_{n,m} e_{n,m}}{f_{n,m}}, \quad (2)$$

where $f_{n,m}$ is the CPU frequency allocated for the task $TK_{n,m}$. Then the local processing energy consumption of the task $TK_{n,m}$ can be given by $E_{n,m}^l = K_n (f_{n,m})^2$, where K_n is a constant characterizing the hardware performance, denoting the energy consumption characteristics during the computation. A larger value of K_n means that the terminal consumes more energy for the same computation. The range of K_n is $[10^{-7}, 10^{-4}]$ [8].

Thus the total energy consumption for AGV n is given by

$$E_n(P_n, f_n, x_n) = \sum_{m=1}^M \{x_{n,m} E_{n,m}^l + (1 - x_{n,m}) E_{n,m}^{comm}\}. \quad (3)$$

The terminal energy efficiency (EE) (bit/J) can be given by $EE_n = \sum_{m=1}^M L_{n,m} / E_n$.

III. LONG TERM ENERGY EFFICIENCY ENABLED JOINT OPTIMIZATION

The objective of this paper is to maximize the AGVs' long term energy efficiency by jointly optimizing computation offloading $x_{n,m}$, power control $P_{n,m}$, and local computational resource allocation $f_{n,m}$ for each AGV. Therefore, the optimization problem can be formulated as

$$\max_{x, P, f} \sum_{n=1}^N EE_n \quad (4)$$

$$x_{n,m} \in \{0, 1\} \quad (5)$$

$$0 \leq f_{n,m}^l \leq F_{n,max} \quad (6)$$

$$0 \leq P_{n,m} \leq P_{n,max} \quad (7)$$

$$t_{n,m} \leq \hat{T}_{n,m} \quad (8)$$

$$e_{n,m} \leq E_{n,m} \quad (9)$$

$$\sum_{m=1}^M e_{n,m} \leq E_n, \quad (10)$$

where (4) is the long term energy efficiency target of the optimization problem, while (5)-(10) are the constraints. Specifically, (5) indicates that the offloading decision is binary, where tasks can either be locally processed or offloaded to the MEC server. (6) limits the local CPU frequency allocated to tasks within the maximum CPU frequency of the AGV. (7) limits the transmission power for task offloading within the maximum transmission power of the AGV. (8) presents the maximum tolerable delay for tasks. Finally, the long term energy efficiency is improved through (9), where $E_{n,m}$ denotes the energy allocated to the task $TK_{n,m}$. and its value will be provided later in this section. $e_{n,m}$ is the actual energy consumption of $TK_{n,m}$. Energy consumption needs to satisfy long-term energy limitations defined as (10) where E_n is

the battery energy of AGV n . Each AGV's total energy consumption must be less than its battery energy. If a MT consumes too much energy early on, it may not be able to complete the remaining tasks.

To improve long-term energy efficiency, it is essential to allocate appropriate energy for each task. Since the tasks on the AGV are similar, the energy consumption of history tasks can be utilized to allocate energy for the current task, given by

$$\frac{TS_{n,m}}{E_n - E_{n,m}^r} = \frac{L_{n,m}}{E_{n,m}}, \quad (11)$$

where $TS_{n,m}$ is the total size of tasks completed previously, $E_{n,m}^r$ is the remaining energy updated by $E_{n,m}^r = E_{n,m-1}^r - e_{n,m-1}$ with $E_{n,1}^r = E_n$, $E_n - E_{n,m}^r$ denotes the total energy consumed for the tasks completed earlier, and $E_{n,0} = E_n$.

IV. EE REWARD-BASED COLLABORATIVE MADRL

A. Definition of EE-CMADRL Components

An EE reward based cooperative MADRL algorithm (EE-CMADRL) is proposed to solve the LTE-JTORA problem (4)-(10), which is NP-hard [1]. As the basic of the algorithm, the definitions of MADRL components, i.e., observation space, action space, and reward function, are given as follows.

a) *Action Space*: In this paper, the action space of agent n is defined as A_n , where $a_{n,m} \in A_n$ denotes the action of agent n at slot m . $a_{n,m}$ is given by

$$a_{n,m} = \{x_{n,m}, P_{n,m}\} \text{ or } \{x_{n,m}, f_{n,m}\}. \quad (12)$$

The joint action of all agents at slot m is denoted as a_m , given by

$$\begin{aligned} a_m &= a_{0,m} \cup \dots \cup a_{n,m} \cup \dots \cup a_{N,m} \\ &= \{x_{0,m}, \dots, x_{n,m}, \dots, x_{N,m}, P_{0,m}, \dots, P_{n,m}, \dots, P_{N,m}\} \\ &\text{ or } \{x_{0,m}, \dots, x_{n,m}, \dots, x_{N,m}, f_{0,m}, \dots, f_{n,m}, \dots, f_{N,m}\} \\ &= \{X_m, P_m\} \text{ or } \{X_m, F_m\}. \end{aligned} \quad (13)$$

b) *Observation Space*: A partially observable Markov decision process model is employed, where each agent makes decisions based on local observations at each slot. Let O_n and $o_{n,m} \in O_n$ denote the observation space of agent n and the observation of agent n at time slot m , respectively. $o_{n,m}$ is given by

$$o_{n,m} = \{g_{n,m-1}, P_{n,m-1}, c_{n,m-1}, f_{n,m-1}\}, \quad (14)$$

where $g_{n,m-1}$ is the channel state information (CSI) between MT n and the BS, $P_{n,m-1}$, $c_{n,m-1}$, and $f_{n,m-1}$ are the transmission power, transmission rate and the CPU frequency of AGV n at slot $m-1$, respectively.

c) *Reward Function*: To maximize the objective function of the optimization problem, i.e., (4), the sum of EE is chosen as the immediate reward, given by $r^m = \sum_{n=1}^N \sum_{m=1}^M EE_{n,m}$.

B. EE-CMADRL

MASC [1] is employed in this paper to accelerate convergence. Centralized Training Distributed Execution (CTDE) framework is employed to address the non-stationarity and dimensionality explosion issues present in other frameworks [1].

In this paper, the deep deterministic policy gradient (DDPG) method is employed to implement each agent. The training process is divided into E episodes, each with S steps. At the beginning of each episode, each agent first interacts with the environment by observing $o_{n,m}$, executing $a_{n,m}$, then receives $r_{n,m}$ and $o_{n,m+1}$ for S consecutive steps, i.e., $m = 1, 2, \dots, S$. As the observation being transitioned from $o_{n,m}$ to $o_{n,m+1}$, the agent n stores the experience $\langle o_{n,m}, a_{n,m}, r_{n,m}, o_{n,m+1} \rangle$ into its experience buffer.

Then, the agents optimize their policies by learning from the collected experiences. Each agent samples Z experiences from its experience buffer, where experience z from agent n can be given by $\langle o_{n,z}, a_{n,z}, r_{n,z}, o_{n,z+1} \rangle$. The target network uses experiences to obtain

$$y_{n,z} = r_{n,z} + \gamma Q_{\omega^-}(o_{n,z+1}, \mu_{n,\theta^-}(o_{n,z+1})), \quad (15)$$

where $\gamma \in [0, 1]$ denotes the discount factor, Q_{ω^-} is the shared target critic network, μ_{n,θ^-} is the target actor network of agent n . Minimize the target loss

$$L = \frac{1}{Z} \sum_{z=1}^Z (y_{n,z} - Q_{\omega}(o_{n,z}, a_{n,z}))^2, \quad (16)$$

to update the shared critic network, where Q_{ω} is the shared critic network. Then, the sampled policy gradient is calculated to update the current agent's actor network, given by

$$\nabla_{n,\theta} J \approx \frac{1}{Z} \sum_{z=1}^Z \nabla_{n,\theta} \mu_{n,\theta}(o_{n,z}) \nabla_{n,\alpha} Q_{\omega}(o_{n,z}, a) \big|_{\alpha=\mu_{n,\theta}(o_z)}, \quad (17)$$

where $\mu_{n,\theta}$ is the actor network of agent n . Finally, the target actor networks and the shared critic network are updated with soft updates, given by

$$\omega^- \leftarrow \tau \omega + (1 - \tau) \omega^-, \quad (18)$$

$$\theta^- \leftarrow \tau \theta + (1 - \tau) \theta^-, \quad (19)$$

where ω^- and θ^- are the parameters of the target networks, ω and θ are the parameters of the current networks, and τ is the soft update rate.

The training phase stops as soon as the E -th episode is finished. Then the trained actor networks can be deployed by the corresponding AGVs to make decisions on joint task offloading and resource allocation. During the execution phase, decisions made by any individual agent do not require any information exchange. Therefore, the proposed EE-CMADRL algorithm is scalable to large-scale networks.

V. PERFORMANCE EVALUATION

In the simulation, we consider an MEC-IIoT system which includes one BS with MEC server and 10 AGVs. The coverage radius of the BS is 200m. For tasks of AGVs, the task size is uniformly distributed at $[5MB, 35MB]$ [8], the maximum tolerant delay of AGVs is 10s, and the number of CPU cycles required to process each bit of data is uniformly distributed at $[1000, 3000]$. The maximum CPU frequency of the MEC server is 6.5e9cycle/s. The CPU frequency of MTs is uniformly distributed at $[0.6e8\text{cycle/s}, 2.5e8\text{cycle/s}]$ [8]. The bandwidth of each AGV is 100 kHz and the noise power is $\sigma^2 = 1 \times 10^{-9}W$ [3]. The path loss between AGV and BS is modeled as $127 + 30 \log(d) \text{ (in dB)}$ [7], where $d(km)$ denotes the distance between AGV and BS. Each AGV's transmission power is uniformly distributed at $[0.01W, 0.2W]$.

To verify the effectiveness of the proposed EE-CMADRL, MADDRG, MASAC, MADDQN and MAAC are selected as baseline.

Fig. 2 presents the policy loss for different algorithms, which can reflect the convergence of algorithms. It is shown that EE-CMADRL and MAAC converge the fastest. However, MAAC suffers from the issue of dimensionality explosion, making it impractical for real-world applications [1]. The CTDE framework and MASC architecture are employed in EE-CMADRL, where multiple agents train a shared critic network, which accelerates the system's convergence. The convergence speed of EE-CMADRL is 71% faster compared to the other three benchmark algorithms.

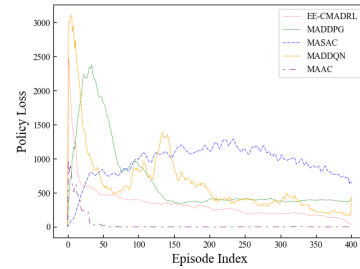


Fig. 2. Policy loss of different MADRL algorithms.

To verify the effectiveness of the proposed LTE-JTORA scheme with EE-CMADRL in long-term energy efficiency, it is compared to short-term energy control (STEC) schemes with different MADRL algorithms.

Fig. 3 shows the total size of completed tasks of LTE-JTORA plus EE-CMADRL algorithm and other benchmark schemes with different battery energy limits. Fig. 3(a) shows the total task size completed by different schemes when AGVs' battery energy is 25J. In Fig. 3(b) and Fig. 3(c), AGVs' battery energy is 50J and 75J, respectively [15]. Fig. 3(d) shows the variation in the total size of tasks completed by different schemes as the battery energy changes. We observe that under different battery energy, the proposed scheme achieves the largest total task completion size. When the

battery energy is 25J, 50J, and 75J, the proposed scheme achieves an average task completion improvements of 93%, 46%, and 31% compared to the benchmark schemes. This is because the proposed scheme considers long term energy efficiency, which helps the AGV complete as many tasks as possible.

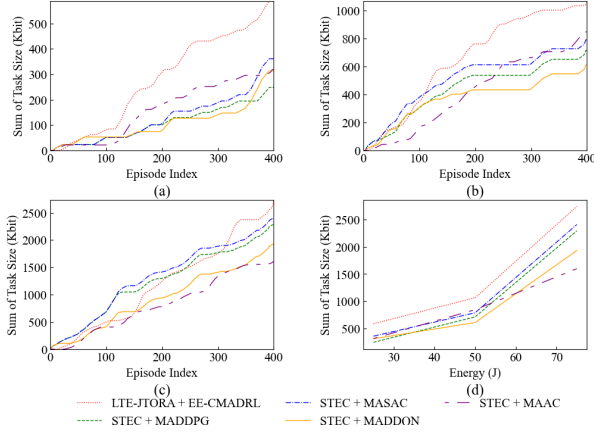


Fig. 3. Total size of completed tasks comparison among different schemes.

Fig. 4 shows the AGVs' energy efficiency of the proposed scheme and other baseline schemes with different battery energy. When the battery energy is 25J, 50J, and 75J, the proposed scheme achieves terminal energy efficiency improvements of 81%, 45%, and 12% compared to the best-performing benchmark schemes. This is because the proposed EE-CMADRL is an energy efficient reward-based MADRL approach, aiming to maximize energy efficiency. Moreover, under the same battery energy, the proposed LTE-JTORA scheme enables AGVs to accomplish more tasks. Therefore, the proposed scheme can improve the AGVs' energy efficiency.

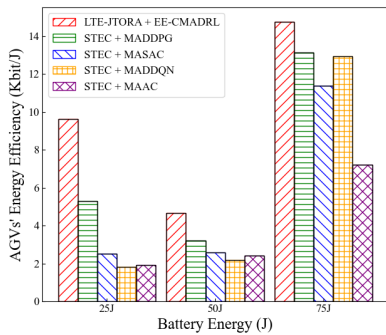


Fig. 4. Long-term energy efficiency comparison among different schemes.

VI. CONCLUSIONS

In this paper, joint task offloading and resource allocation in MEC-IIoT system have been studied to maximize AGVs' long term energy efficiency. To address this joint optimization problem, we proposed a long-term energy control enabled joint

optimization of task offloading and resource allocation scheme (LTE-JTORA) and an energy efficient (EE) reward-based collaborative multi-agent DRL approach (EE-CMADRL). The proposed EE-CMADRL algorithm demonstrated superior performance in convergence. The superior performance of the proposed LTE-JTORA scheme with EE-CMADRL in long term energy efficiency have been demonstrated by the simulation results.

REFERENCES

- [1] Y. Xiao, Y. Song and J. Liu, "Collaborative Multi-Agent Deep Reinforcement Learning for Energy-Efficient Resource Allocation in Heterogeneous Mobile Edge Computing Networks," *IEEE Transactions on Wireless Communications*, vol. 23, no. 6, pp. 6653-6668, June 2024.
- [2] S. Zarandi and H. Tabassum, "Federated Double Deep Q-learning for Joint Delay and Energy Minimization in IoT networks," 2021 IEEE International Conference on Communications Workshops (ICC Workshops), Montreal, QC, Canada, 2021, pp. 1-6.
- [3] R. Zhang, C. Pan, Y. Wang, Y. Yao and X. Li, "Federated Deep Reinforcement Learning for Multimedia Task Offloading and Resource Allocation in MEC Networks," *IEEE Transactions on Communications*, vol. E107-B, no. 6, pp. 446-457, June 2024.
- [4] F. Liang, W. Yu, X. Liu, D. Griffith and N. Golmie, "Toward Deep Q-Network-Based Resource Allocation in Industrial Internet of Things," *IEEE Internet of Things Journal*, vol. 9, no. 12, pp. 9138-9150, 15 June 2022.
- [5] Y. Xu, L. Zheng, X. Wu, Y. Tang, W. Liu and D. Sun, "Energy-Efficient Resource Allocation for V2X Communications," *IEEE Internet of Things Journal*, vol. 11, no. 18, pp. 30014-30026, 15 Sept. 2024.
- [6] P. Qin et al., "Joint Resource Allocation and UAV Trajectory Design for D2D-Assisted Energy-Efficient Air-Ground Integrated Caching Network," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 11, pp. 17558-17571, Nov. 2024.
- [7] Y. Chen, Z. Liu, Y. Zhang, Y. Wu, X. Chen and L. Zhao, "Deep Reinforcement Learning-Based Dynamic Resource Management for Mobile Edge Computing in Industrial Internet of Things," *IEEE Transactions on Industrial Informatics*, vol. 17, no. 7, pp. 4925-4934, July 2021.
- [8] Y. Gong, H. Yao, J. Wang, M. Li and S. Guo, "Edge Intelligence-Driven Joint Offloading and Resource Allocation for Future 6G Industrial Internet of Things," *IEEE Transactions on Network Science and Engineering*, vol. 11, no. 6, pp. 5644-5655, Nov.-Dec. 2024.
- [9] R. Xiong, C. Zhang, H. Zeng, X. Yi, L. Li and P. Wang, "Reducing Power Consumption for Autonomous Ground Vehicles via Resource Allocation Based on Road Segmentation in V2X-MEC With Resource Constraints," *IEEE Transactions on Vehicular Technology*, vol. 71, no. 6, pp. 6397-6409, June 2022.
- [10] Y. Wu, X. Zhu, J. Fei and H. Xu, "A Novel Joint Optimization Method of Multi-Agent Task Offloading and Resource Scheduling for Mobile Inspection Service in Smart Factory," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 6, pp. 8563-8575, June 2024.
- [11] Z. Yang, C. Pan, K. Wang and M. Shikh-Bahaei, "Energy Efficient Resource Allocation in UAV-Enabled Mobile Edge Computing Networks," *IEEE Transactions on Wireless Communications*, vol. 18, no. 9, pp. 4576-4589, Sept. 2019.
- [12] J. Feng, Q. Pei, F. R. Yu, X. Chu, J. Du and L. Zhu, "Dynamic Network Slicing and Resource Allocation in Mobile Edge Computing Systems," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 7, pp. 7863-7878, July 2020.
- [13] D. Wang, H. Qin, B. Song, X. Du, and M. Guizani, "Resource allocation in information-centric wireless networking with D2D-enabled MEC: A deep reinforcement learning approach," *IEEE Access*, vol. 7, pp. 114 935-114 944, 2019.
- [14] L. Kraemer and B. Banerjee, "Multi-agent reinforcement learning as a rehearsal for decentralized planning," *Neurocomputing*, vol. 190, pp. 82-94, 2016.
- [15] J. Xu and H. Wang, "Client Selection and Bandwidth Allocation in Wireless Federated Learning Networks: A Long-Term Perspective," *IEEE Transactions on Wireless Communications*, vol. 20, no. 2, pp. 1188-1200, Feb. 2021.